

Optimization of Ring Shell Structures with Multiple Flexoelectric Actuators Based on Sequential Quadratic Programming

Abstract

This study presents a multi-channel design and optimization framework for active vibration control of ring shell structures using flexoelectric actuators. The novelty of this work is three-fold: (i) a coupled electromechanical dynamic model that captures both membrane and bending responses of a freely floating ring under localized flexoelectric actuation; (ii) a practical actuation mechanism based on AFM-probe-generated nonuniform electric fields to produce highly localized converse-flexoelectric driving forces; and (iii) a multi-objective Sequential Quadratic Programming (SQP) procedure that simultaneously optimizes actuator positions and drive voltages to minimize a physically motivated roundness function measuring deviation from circularity. Numerical case studies for a ring (radius 0.05 m, thickness 0.001 m; flexoelectric patch thickness 50 μm ; AFM tip radius 50 nm; actuation voltages up to 10 V) demonstrate that the SQP routine robustly identifies actuator configurations that cancel overall vibration across multiple excited modes, yielding near-zero residual deformation where single-channel schemes fail. We further show that the optimal actuator distributions obey cyclotomic-polynomial symmetry and that control benefit saturates as the actuator count increases (computational cost also rises), indicating an effective practical range for actuator number. The approach offers a systematic route for precision deformation control in ring-shaped micro- and nano-devices and informs design choices for multi-actuator flexoelectric systems.

Keywords: Flexoelectric Effect· Vibration control· Sequential Quadratic Programming· Ring Shell Structures

1. Introduction

The inherent flexibility of ring shell structures and their sensitivity to dynamic excitations make them particularly susceptible to disturbance-induced vibrations, thereby attracting widespread research interest. The ring structure is also one of the most common structural models in engineering applications, such as fixed bridge structures(Liu et al. 2024), pipeline systems(Olso and Kyriakides 2003), storage tanks(Kameshwar and Padgett 2019) and mechanical transmission systems(Fan et al. 2020). Traditional passive methods are often inadequate in dynamic settings, prompting the development of active strategies such as embedded or surface-mounted actuators for precise control of ring shell deformations and vibrations.

In recent decades, there has been extensive research on precise actuation and vibration control of various structures(Tzou et al. 2017a;Ellison et al. 2001;Lin et al. 2009;Zhai et al. ;Hoang and Thanh 2025;Vu-Bac et al. 2018;Vu-Bac et al. 2019;Vu-Bac et al. 2022). Among them, piezoelectric materials have been widely used in engineering applications as sensors and actuators based on the piezoelectric effect, achieving precise actuation or vibration control(Gao et al. 2020;Mohith et al. 2020;Kundalwal and Shingare 2020). In comparison with the piezoelectric effect, the flexoelectric effect exists in all dielectric solids, thus attracting more researchers' interest(Zubko et al. 2013;Yudin and Tagantsev 2013;Deng et al. 2020;Ghasemi et al. 2020;Ghasemi et al. 2017;Ghasemi et al. 2018). Gradient coupling is an intrinsic characteristic of the flexoelectric effect(Ma et al. 2024;Sun et al. 2023). The flexoelectric effect can be divided into direct flexoelectric effect and converse flexoelectric effect(Mao et al. 2016). Smart materials based on the converse flexoelectric effect have emerged as a research hotspot-as alternatives to piezoelectric materials-due to their capacity for more precise microscale actuation(Bhaskar et al. 2016;Hu et al. 2013;Hu et al. 2014;Han et al. 2023). Building on this, the present study analyzes the microscopic actuation behavior of flexoelectric actuator patches on elastic ring shells. Early studies by Hu and Tzou(Hu and Tzou 2010) and Li(Li et al. 2015) demonstrated the feasibility of segmented patch designs by affixing flexoelectric sensor layers to ring and conical shells, successfully capturing modal signal distributions. Subsequently, the applications of flexoelectric actuators for vibration control and actuation across various structural configurations were investigated(Tzou and Zhang 2016;Zhang et al. 2019;Deng et al. 2015).

However, for non-planar structures such as thin shells, complex vibration patterns and higher modes necessitate multi-channel control, and thus call for optimized multi-channel actuator design. Moreover, no previous studies have addressed multi-channel design for vibration control of ring shells. This challenge is inherently a nonlinear programming problem-its objective and constraints include nonlinear terms-and thus demands specialized solution techniques. Convex optimization theory underpins many such approaches, with gradient-based methods proving particularly effective(Nesterov 2015). In this work, we employ Sequential Quadratic Programming (SQP), a powerful gradient-based algorithm that has been widely applied across diverse fields, including machining process optimization(Chen and Ding

2023), mineral extraction(Nguyen et al. 2023;Nguyen et al. 2024), power transmission(Hasan et al. 2024), and aerospace guidance(Hong et al. 2021).

This study employs an AFM probe-generated nonuniform electric field for precision actuation. An analysis of the influence of a single flexoelectric patch on the displacement of a specific point in a ring structure under different structural parameters and actuator parameters was first conducted. Based on these results, parameters with significant effects were selected, and then the SQP algorithm was used to analyze the optimal arrangement of multiple actuators to achieve optimal driving and control effects.

2. Physical Model

2.1. The flexoelectric dynamical modeling of ring shell structures

As mentioned earlier, using multiple flexoelectric actuators can alleviate stress concentration issues in the ring structures. To achieve this effect, an inhomogeneous electric field is generated within the flexoelectric patches, followed by the generation of control forces through the flexoelectric actuators. Moreover, the ring is modeled using Kirchhoff-Love thin shell theory, which assumes: (i) Linear strain-displacement relations; (ii) Negligible transverse shear deformations $\varepsilon_{3\psi} = \varepsilon_{3s}$; (iii) Stress normal to the mid-surface $\sigma_{33} \approx 0$; These simplifications are valid for thin shells where $h/R \ll 1$ (Soedel 2004).

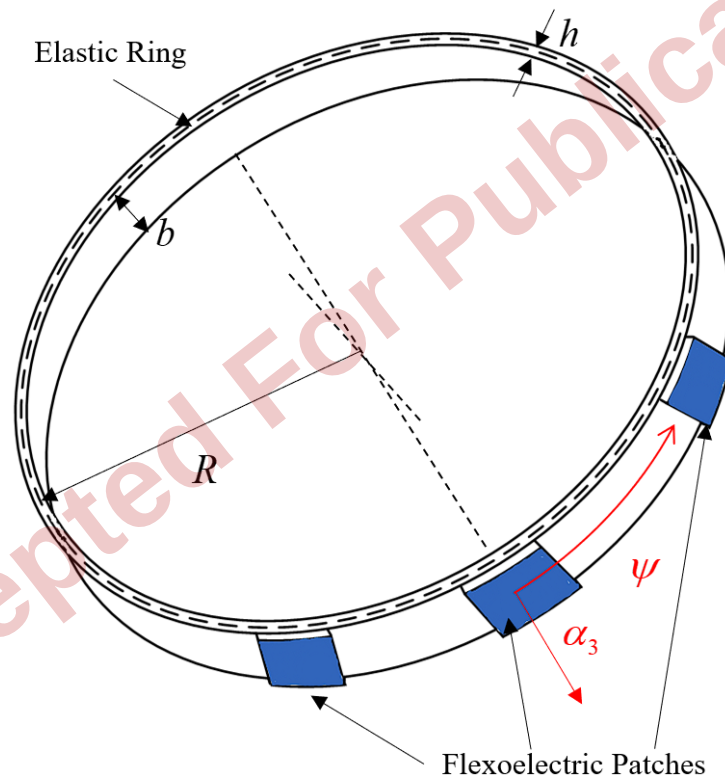


Fig. 1 Schematic diagram of the elastic ring

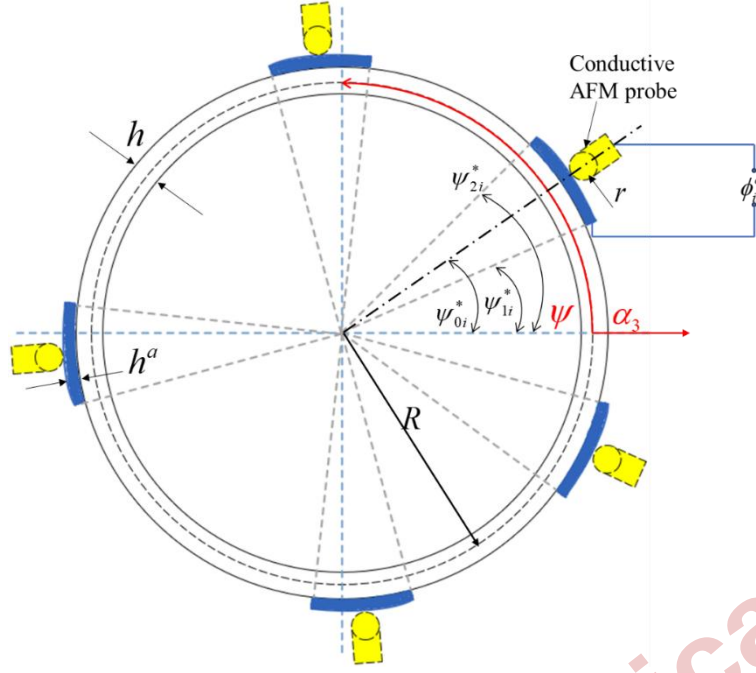


Fig. 2 A ring model of multiple flexoelectric actuators

Figures 1 and 2 depict schematic representations of ring structures equipped with multiple flexoelectric actuators, with α_3 and ψ defining the radial and circumferential directions in an orthogonal coordinate system, respectively; R is the neutral surface radius of the ring; h is the ring thickness, and b is the ring width. Regarding the actuators, the dimensional attributes of Atomic Force Microscope (AFM) probes are characterized by the probe radius r , h^a represents the thickness of each flexoelectric actuator, and ψ_{0i} , ψ_{1i} , and ψ_{2i} respectively represent the AFM application position, start position, and end position of each flexoelectric patch. For analytical simplification, it is assumed that all flexoelectric patches have identical dimensions. It is also assumed that the probe radius r , flexoelectric patch thickness h^a , and ring thickness h adhere to the following relationship, namely, $r \ll h_a \ll h$, the flexoelectric patch is much thinner than the ring, thus neglecting its stiffness/mass effect on the ring dynamics. Therefore, by simplifying the dynamic equations of the ring from those of the double-curvature shell, considering mechanical and actuation forces, the following equations can be obtained (Tzou et al. 2017b):

$$\rho h \frac{\partial^2 u_\psi}{\partial t^2} - \frac{1}{R} \frac{\partial N_{\psi\psi}^m}{\partial \psi} - \frac{1}{R^2} \frac{\partial M_{\psi\psi}^m}{\partial \psi} = -\frac{1}{R} \frac{\partial N_{\psi\psi}^a}{\partial \psi} - \frac{1}{R^2} \frac{\partial M_{\psi\psi}^a}{\partial \psi} \quad (1)$$

$$\rho h \frac{\partial^2 u_3}{\partial t^2} + \frac{N_{\psi\psi}^m}{R} - \frac{1}{R^2} \frac{\partial^2 M_{\psi\psi}^m}{\partial \psi^2} = \frac{N_{\psi\psi}^a}{R} - \frac{1}{R^2} \frac{\partial^2 M_{\psi\psi}^a}{\partial \psi^2} + F_3 \quad (2)$$

Where ρ is the mass density of the ring; u_3 is the transverse displacement; u_ψ is the circumferential displacement; F_3 is the transversely distributed mechanical force; $N_{\psi\psi}^a$ and $M_{\psi\psi}^a$, respectively, the elastic membrane force and bending moment per unit width, with the superscript a denoting the actuator-induced component. Similarly, $N_{\psi\psi}^m$ and $M_{\psi\psi}^m$ are, respectively, the elastic membrane force and bending moment per unit width, where the superscript m denotes the mechanical component. Circumferential and transverse motions are coupled, with transverse oscillation typically dominating. Combining the actuator-induced components in the ring equation, Love's control operator can be expressed as:

$$L_\psi^a = -\frac{1}{R} \frac{\partial N_{\psi\psi}^a}{\partial \psi} - \frac{1}{R^2} \frac{\partial M_{\psi\psi}^a}{\partial \psi} \quad (3)$$

$$L_3^a = \frac{N_{\psi\psi}^a}{R} - \frac{1}{R^2} \frac{\partial^2 M_{\psi\psi}^a}{\partial \psi^2} \quad (4)$$

L_ψ^a and L_3^a represent the Love's actuation or control operators, which are induced by the flexoelectric actuators in the circumferential and radial directions, respectively. The ring structure requires periodic boundary conditions, i.e., $u_\psi(\psi + 2\pi) = u_\psi(\psi)$ and $u_3(\psi + 2\pi) = u_3(\psi)$ (Vu-Bac et al. 2015). Through analysis, it can be seen that the dynamic displacement of the ring can be expressed by the modal expansion method, which represents the dynamic displacement of the ring as the sum of the modal coefficients of all modes multiplied by the corresponding modal functions.

$$\begin{aligned} u_\psi(\psi, t) &= \sum_{k=2}^{\infty} \eta_k(t) U_{\psi k}(\psi, t) \\ u_3(\psi, t) &= \sum_{k=2}^{\infty} \eta_k(t) U_{3k}(\psi, t) \end{aligned} \quad (5)$$

where k represents the mode number; η_k is the modal participation factor, which is time-dependent; U_{3k} and $U_{\psi k}$ are, respectively, the transverse and circumferential mode shape functions of the k th mode, which are spatially dependent. The boundary conditions of the ring are set to a freely floating state, observing that $k=0$ is a breathing mode; $k=1$ is a rigid body mode, and when $k \geq 2$, a set of transverse and circumferential modes occur. The mode shape functions for the circumferential and transverse directions are respectively represented as:

$$\begin{aligned} U_{\psi k} &= A_k \sin(k\psi - \varphi) \\ U_{3k} &= B_k \cos(k\psi - \varphi) \end{aligned} \quad (6)$$

where A_k and B_k are the modal amplitudes, and φ is an arbitrary phase angle representing arbitrary ring orientation. For simplicity, the authors assume that the value of φ is 0 (Shundi 2014). Substituting the modal expansions, i.e., Eq. (5a) and (5b), into the control equations of the ring, taking note of the orthogonality of the mode shape functions, and introducing modal damping, we obtain the independent modal equations:

$$\ddot{\eta}_k + 2\zeta_k \omega_k \dot{\eta}_k + \omega_k^2 \eta_k = \hat{F}_k \quad (7)$$

In which, ω_k represents the k th natural frequency; ζ_k is the modal damping ratio defined by an equivalent damping constant c and the natural frequency ω_k , that is, $\zeta_k = c / (2\rho h \omega_k)$; $\hat{F}_k$ denotes the k th modal force. For each mode $k \geq 2$, there are two component natural frequencies, namely ω_{k1} and ω_{k2} , representing transverse and circumferential vibrations, respectively. At lower k , i.e., $k \leq 10$, the modal amplitudes of transverse and circumferential component modes are coupled. By substituting the mode shape functions into the control equations of the ring, the relationship between the modal amplitude A_k of the circumferential mode and that of the transverse mode B_k can be obtained. For vibrations at frequencies ω_{k1} and $k \leq 10$, the ratio of the amplitude of the transverse mode B_k to that of the circumferential mode A_k is proportional to k , i.e., $(B_k / A_k) \approx -k$ (Soedel 2004).

For typical rings, $\omega_{k2} \gg \omega_{k1}$; therefore, circumferential vibrations are not considered, and only transverse vibrations ω_{k1} are considered for subsequent vibration control. The modal force can be divided into two components: the modal force induced by mechanical force $\hat{F}_k^m$ and that induced by flexoelectric actuators $\hat{F}_k^a$, namely, $\hat{F}_k = \hat{F}_k^m + \hat{F}_k^a$. Let the external mechanical force in the transverse direction be represented by F_3 . The modal force components $\hat{F}_k^m$ and $\hat{F}_k^a$ are defined as:

$$\begin{aligned} \hat{F}_k^m &= \frac{1}{\rho h N_k} \int_0^{2\pi} F_3 U_{3k} R d\psi \\ \hat{F}_k^a &= \frac{1}{\rho h N_k} \int_{\psi_1^*}^{\psi_2^*} (L_\psi^a U_{\psi k} + L_3^a U_{3k}) R d\psi \end{aligned} \quad (8)$$

where, $N_k = \int_0^{2\pi} (U_{\psi k}^2 + U_{3k}^2) R d\psi$. It is worth noting that the flexoelectric patch extends

counterclockwise from φ_{n1} to φ_{n2} . When the mechanical force F_3 and the excitation voltage ϕ^a can be represented as, i.e., $F_3 = F_3^* e^{j\omega t}$, $\phi^a(t) = \phi^a e^{j\omega t}$, then the steady-state modal response is also harmonic, $\hat{F}_k = \hat{F}_k^* e^{j\omega t}$, where $\hat{F}_k^*$ represents the magnitude of the k th modal force. By performing the Laplace transform, the modal Eq. (7) can be solved to obtain the steady-state modal response.

$$\begin{aligned} \eta_k(t) &= \frac{\hat{F}_k^*}{(\omega_k^2 - \omega^2) + 2j\zeta_k \omega_k \omega} e^{j\omega t} \\ &= \frac{\hat{F}_k^*}{\omega_k^2 \sqrt{\left(1 - \frac{\omega^2}{\omega_k^2}\right)^2 + 4\zeta_k^2 \left(\frac{\omega}{\omega_k}\right)^2}} e^{j(\omega t - \varphi^*)} \end{aligned} \quad (9)$$

where φ^* is the phase angle expressed as $\varphi^* = \arctan\left[\left(2\zeta_k(\omega/\omega_k)\right) / \left(1 - (\omega^2/\omega_k^2)\right)\right]$.

The orientation of the ring is determined by the distribution of external loads. Substituting the mode vibration function into the electrical-machine-ring equation yields (Tzou 1993):

$$\begin{pmatrix} \rho h \omega_k^2 - \frac{k^2 D}{R^4} - \frac{k^2 K}{R^2} & -\frac{k^3 D}{R^4} - \frac{kK}{R^2} \\ -\frac{k^3 D}{R^4} - \frac{kK}{R^2} & \rho h \omega_k^2 - \frac{k^4 D}{R^4} - \frac{K}{R^2} \end{pmatrix} \begin{bmatrix} A_n \\ B_n \end{bmatrix} = 0 \quad (10)$$

For non-trivial solutions, the modal amplitudes $\begin{bmatrix} A_n \\ B_n \end{bmatrix} \neq 0$, therefore,

$$\det \left(\begin{pmatrix} \rho h \omega_k^2 - \frac{k^2 D}{R^4} - \frac{k^2 K}{R^2} & -\frac{k^3 D}{R^4} - \frac{kK}{R^2} \\ -\frac{k^3 D}{R^4} - \frac{kK}{R^2} & \rho h \omega_k^2 - \frac{k^4 D}{R^4} - \frac{K}{R^2} \end{pmatrix} \right) = 0 \quad (11)$$

Expanding the determinant and simplifying yields the characteristic equation:

$$\omega_n^4 - K_1 \omega_n^2 + K_2 = 0 \text{ s} \quad (12)$$

where

$$K_1 = \frac{k^2 + 1}{R^2 \rho h} \left(\frac{k^2 D}{R^2} + K \right) \quad (13)$$

$$K_2 = \frac{k^2 (k^2 - 1)^2}{R^6 (\rho h)} DK \quad (14)$$

where, the membrane stiffness K and bending stiffness D are defined as

$$K = \frac{Yh}{1 - \mu^2} \quad (15)$$

$$D = \frac{Yh^3}{12(1 - \mu^2)} \quad (16)$$

In the above equations, Y represents the Young's modulus, and μ represents the Poisson's ratio.

2.2. Vibration control with multiple piezoelectric actuators

The flexoelectric effect can be categorized into two distinct phenomena: the direct flexoelectric effect and the converse flexoelectric effect. The former refers to the generation of polarization in dielectric materials when subjected to non-uniform mechanical deformation, specifically strain gradients. Conversely, the latter describes the mechanical strain induced in dielectric substances under the influence of non-uniform electric fields, characterized by electric field gradients.

In this study, the non-uniform electric field generated by the AFM probe can be expressed as (Tzou et al. 2017b):

$$\phi = \frac{\phi^a r}{\sqrt{(\alpha_3 + R)^2 \sin^2(\psi - \psi_{0i}^*) + \left[r + \frac{h}{2} + h^a - (\alpha_3 + R) \cos(\psi - \psi_{0i}^*) + R \right]^2}} \quad (17)$$

In this study, the electric field in the natural environment is almost 0, so it can be ignored. Moreover, the actuators do not interfere with each other, so only the electric field generated by the corresponding AFM probe needs to be considered. Therefore, the electric field intensity generated by the AFM probe can be obtained by calculating the partial derivative:

$$E_3 = -\frac{\partial \phi}{\partial \alpha_3} = \phi^a r \left\{ (\alpha_3 + R) \sin^2(\psi - \psi_{0i}^*) - \left[r + \frac{h}{2} + h^a - (\alpha_3 + R) \cos(\psi - \psi_{0i}^*) + R \right] \cos(\psi - \psi_{0i}^*) \right\} \times \left\{ (\alpha_3 + R)^2 \sin^2(\psi - \psi_{0i}^*) + \left[r + \frac{h}{2} + h^a - (\alpha_3 + R) \cos(\psi - \psi_{0i}^*) + R \right]^2 \right\}^{\frac{3}{2}} \quad (18)$$

The circumferential stress induced by the electric field can be defined from the converse flexoelectric equation:

$$\begin{aligned} T_{\psi\psi}^a &= \pi_{12} \frac{\partial E_3}{\partial \alpha_3} \\ &= \frac{\pi_{12} \phi^a r}{\left\{ (\alpha_3 + R)^2 \sin^2(\psi - \psi_{0i}^*) + \left[r + \frac{h}{2} + h^a - (\alpha_3 + R) \cos(\psi - \psi_{0i}^*) + R \right]^2 \right\}^{\frac{3}{2}}} \\ &\quad - \frac{\frac{3}{2} \pi_{12} \phi^a r}{\left\{ (\alpha_3 + R)^2 \sin^2(\psi - \psi_{0i}^*) + \left[r + \frac{h}{2} + h^a - (\alpha_3 + R) \cos(\psi - \psi_{0i}^*) + R \right]^2 \right\}^{\frac{5}{2}}} \\ &\quad \times \left\{ (\alpha_3 + R) \sin^2(\psi - \psi_{0i}^*) - \left[r + \frac{h}{2} + h^a - (\alpha_3 + R) \cos(\psi - \psi_{0i}^*) + R \right] \cos(\psi - \psi_{0i}^*) \right\} \\ &\quad \times \left\{ 2(\alpha_3 + R) \sin^2(\psi - \psi_{0i}^*) - 2 \left[r + \frac{h}{2} + h^a - (\alpha_3 + R) \cos(\psi - \psi_{0i}^*) + R \right] \cos(\psi - \psi_{0i}^*) \right\} \end{aligned} \quad (19)$$

$T_{\psi\psi}^a$ represents the circumferential stress in the flexoelectric actuator. The control force of the flexoelectric membrane can be obtained by integrating the stress along the actuator thickness:

$$N_{\psi\psi}^a = \int_{\frac{h}{2}}^{\frac{h}{2} + h^a} T_{\psi\psi}^a d\alpha_3 \quad (20)$$

The flexoelectric control moment is calculated by integrating the product of the actuation stress and the corresponding moment arm, which is the distance between the local point and the ring's neutral layer.

$$M_{\psi\psi}^a = \int_{\frac{h}{2}}^{\frac{h}{2} + h^a} \alpha_3 T_{\psi\psi}^a d\alpha_3 \quad (21)$$

Substituting the above equations into Eq.(9) yields the expression for the displacement of the flexoelectric ring.

$$u_{3i} = \sum_{k=2}^{\infty} \frac{\hat{F}_{ki}^a}{\omega_k^2 \sqrt{\left(1 - \frac{\omega_j^2}{\omega_k^2}\right)^2 + 4\zeta_k^2 \left(\frac{\omega_j}{\omega_k}\right)^2}} U_{3k}(\psi, t) \quad (22)$$

We define u_{3i} as the displacement induced on the ring structure by the i th actuator under a control voltage of frequency ω_j . This displacement is given by the superposition (sum) of that actuator's responses across all modes k .

Next, we define the total control displacement u_3 , which consists of two components: the first is the total displacement generated collectively by the n actuators, i.e., the sum of all u_{3i} ; the second is the displacement induced by an external disturbance force $\hat{F}_3^m$ with excitation frequency ω_m .

$$u_3 = \sum_{i=1}^n u_{3i} + \sum_{k=2}^{\infty} \frac{\hat{F}_3^m}{\omega_k^2 \sqrt{\left(1 - \frac{\omega_j^2}{\omega_k^2}\right)^2 + 4\zeta_k^2 \left(\frac{\omega_j}{\omega_k}\right)^2}} U_{3k}(\psi, t) \quad (23)$$

2.3. Vibration control optimization based on sqp

By the definition of the roundness function and as shown in **Fig.3**, it is clear that the minimum value of the function indicates the maximum vibration amplitude, while the maximum value of the function, which is 1, means that the ring has almost no vibration. Therefore, the roundness function serves well as an optimization target. These will be analyzed in more detail in the following sections.

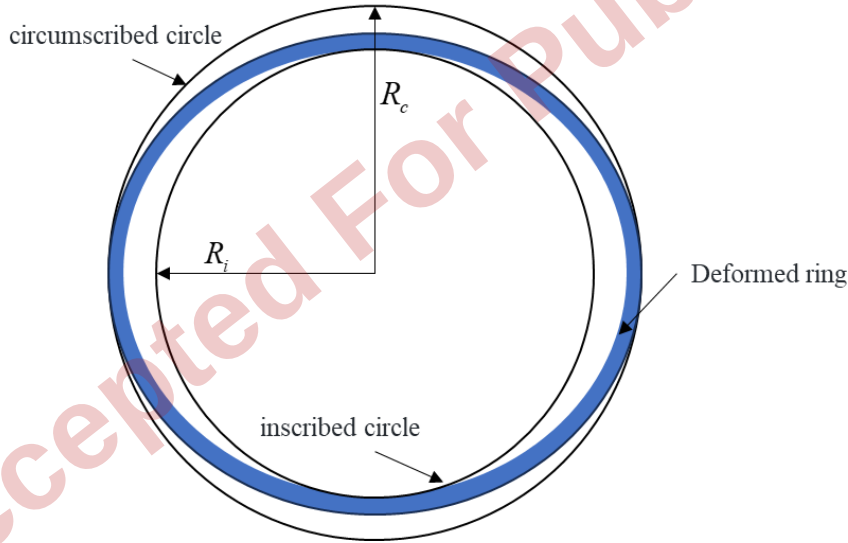


Fig.3 Definition of the roundness function using the mode 2 as an example

Assuming that the objective function $\Gamma(\mathbf{x}) = R_i / R_c$ satisfies the Karush-Kuhn-Tucker (KKT) conditions, we consider $g(\mathbf{x}) = \bigcap_{i=1}^{n-1} \bigcap_{j=i+1}^n \{|\varphi_i - \varphi_{j+1}| \geq (L/R)\}$ as active constraints, which means

$g_i(\mathbf{x}_k) \geq 0 \Rightarrow g_i^*(\mathbf{x}_{k+1}) = 0$. The function $g(\mathbf{x})$ means that no two piezoelectric patches will overlap.

$g^*(\mathbf{x})$ denotes the derivative of $g(\mathbf{x})$. The active constraints extracted from the constraint set are added to the working set, with equality constraints always remaining in the working set. This set evolves continuously with each iteration.

The Lagrange function is constructed as follows:

$$L(\mathbf{x}, \boldsymbol{\mu}) = \Gamma(\mathbf{x}) + \boldsymbol{\mu} g^*(\mathbf{x}) \quad (24)$$

here, $\boldsymbol{\mu}$ denotes the coefficient of $g^*(\mathbf{x})$. Since the function $L(\mathbf{x}, \boldsymbol{\mu})$ satisfies the KKT conditions, it in particular meets the stationarity condition — that is, the gradient of $L(\mathbf{x}, \boldsymbol{\mu})$ vanishes at an extremum. The, the gradient of the function $L(\mathbf{x}, \boldsymbol{\mu})$ is

$$\nabla L = \begin{bmatrix} \frac{\partial L}{\partial \mathbf{x}} \\ \frac{\partial L}{\partial \boldsymbol{\mu}} \end{bmatrix} = \begin{bmatrix} \Gamma(\mathbf{x}) + \boldsymbol{\mu} \nabla g^*(\mathbf{x}) \\ g^*(\mathbf{x}) \end{bmatrix} = \mathbf{0} \quad (25)$$

Expanding from the Newton-Raphson iteration method for general equations to multiple dimensions yields:

$$\mathbf{x}_{k+1} = \mathbf{x}_k - [H(\Gamma(\mathbf{x}_k))]^{-1} \nabla \Gamma(\mathbf{x}_k) \quad (26)$$

here, $H(\Gamma(\mathbf{x}_k))$ denotes the Hessian matrix of the $\Gamma(\mathbf{x}_k)$. From the Taylor expansion of multivariable functions, we can obtain:

$$\Gamma(\mathbf{x}) = \Gamma(\mathbf{x}_k) + [\nabla \Gamma(\mathbf{x}_k)]^T \mathbf{d}_k + \frac{1}{2} \mathbf{d}_k^T \nabla^2 \Gamma(\mathbf{x}_k) \mathbf{d}_k \quad (27)$$

Substituting Eq.(28) into formula Eq.(31) yields the Lagrange equation of the objective function.

$$\Gamma(\mathbf{x}_k) + [\nabla \Gamma(\mathbf{x}_k)]^T \mathbf{d}_k + \frac{1}{2} \mathbf{d}_k^T \nabla^2 \Gamma(\mathbf{x}_k) \mathbf{d}_k + \boldsymbol{\mu} [g^*(\mathbf{x}_k) + [\nabla g^*(\mathbf{x}_k)]^T \mathbf{d}_k] = 0 \quad (28)$$

here, $\mathbf{d}_k = \mathbf{x}_{k+1} - \mathbf{x}_k$. Taking partial derivatives of the objective function yields the following system of equations:

$$\begin{cases} \nabla \Gamma(\mathbf{x}_k) + \nabla^2 \Gamma(\mathbf{x}_k) \mathbf{d}_k + \boldsymbol{\mu} \nabla g^*(\mathbf{x}_k) = 0 \\ g^*(\mathbf{x}_k) + [\nabla g^*(\mathbf{x}_k)]^T \mathbf{d}_k = 0 \end{cases} \quad (29)$$

Rewriting it in matrix form, we have:

$$\begin{bmatrix} \mathbf{d}_k \\ \boldsymbol{\mu} \end{bmatrix} = - \begin{bmatrix} H[\Gamma(\mathbf{x}_k)] & \nabla g^*(\mathbf{x}_k) \\ [\nabla g^*(\mathbf{x}_k)]^T & \mathbf{0} \end{bmatrix} \begin{bmatrix} \nabla \Gamma(\mathbf{x}_k) \\ g^*(\mathbf{x}_k) \end{bmatrix} \quad (30)$$

Then, get the expression for $\mathbf{d}_k$ and $\boldsymbol{\mu}$:

$$\begin{aligned} \mathbf{d}_k &= - (H[\Gamma(\mathbf{x}_k)] \cdot \nabla \Gamma(\mathbf{x}_k) + \nabla g^*(\mathbf{x}_k) \cdot g^*(\mathbf{x}_k)) \\ \boldsymbol{\mu} &= [\nabla g^*(\mathbf{x}_k)]^T \cdot \nabla \Gamma(\mathbf{x}_k) \end{aligned} \quad (31)$$

Rewriting it in matrix form, we have:

$$\mathbf{d}_k = \begin{bmatrix} \varphi_1^{k+1} - \varphi_1^k \\ \phi_1^{k+1} - \phi_1^k \\ \varphi_2^{k+1} - \varphi_2^k \\ \phi_2^{k+1} - \phi_2^k \\ \varphi_3^{k+1} - \varphi_3^k \\ \phi_3^{k+1} - \phi_3^k \\ \varphi_4^{k+1} - \varphi_4^k \\ \vdots \\ \varphi_n^{k+1} - \varphi_n^k \\ \phi_n^{k+1} - \phi_n^k \end{bmatrix} = - \begin{pmatrix} \frac{\partial^2 \Gamma}{\partial \varphi_1^2} \cdot \frac{\partial \Gamma}{\partial \varphi_1} + \frac{\partial^2 \Gamma}{\partial \varphi_1 \partial \phi_1} \cdot \frac{\partial \Gamma}{\partial \phi_1} + \dots + \frac{\partial^2 \Gamma}{\partial \varphi_1 \partial \phi_n} \cdot \frac{\partial \Gamma}{\partial \phi_n} + 2\varphi_1^k - \varphi_2^k - \varphi_n^k + 2\pi \\ \frac{\partial^2 \Gamma}{\partial \phi_1 \partial \varphi_1} \cdot \frac{\partial \Gamma}{\partial \varphi_1} + \frac{\partial^2 \Gamma}{\partial \phi_1^2} \cdot \frac{\partial \Gamma}{\partial \phi_1} + \dots + \frac{\partial^2 \Gamma}{\partial \phi_1 \partial \phi_n} \cdot \frac{\partial \Gamma}{\partial \phi_n} \\ \frac{\partial^2 \Gamma}{\partial \varphi_2 \partial \varphi_1} \cdot \frac{\partial \Gamma}{\partial \varphi_1} + \frac{\partial^2 \Gamma}{\partial \varphi_2 \partial \phi_1} \cdot \frac{\partial \Gamma}{\partial \phi_1} + \dots + \frac{\partial^2 \Gamma}{\partial \varphi_2 \partial \phi_n} \cdot \frac{\partial \Gamma}{\partial \phi_n} + 2\varphi_3^k - \varphi_2^k - \varphi_1^k \\ \vdots \\ \frac{\partial^2 \Gamma}{\partial \varphi_n \partial \varphi_1} \cdot \frac{\partial \Gamma}{\partial \varphi_1} + \frac{\partial^2 \Gamma}{\partial \varphi_n \partial \phi_1} \cdot \frac{\partial \Gamma}{\partial \phi_1} + \dots + \frac{\partial^2 \Gamma}{\partial \varphi_n \partial \phi_n} \cdot \frac{\partial \Gamma}{\partial \phi_n} + 2\varphi_n^k - \varphi_1^k - \varphi_{n-1}^k - 2\pi \\ \frac{\partial^2 \Gamma}{\partial \phi_n \partial \varphi_1} \cdot \frac{\partial \Gamma}{\partial \varphi_1} + \frac{\partial^2 \Gamma}{\partial \phi_n \partial \phi_1} \cdot \frac{\partial \Gamma}{\partial \phi_1} + \dots + \frac{\partial^2 \Gamma}{\partial \phi_n^2} \cdot \frac{\partial \Gamma}{\partial \phi_n} \end{pmatrix} \quad (32)$$

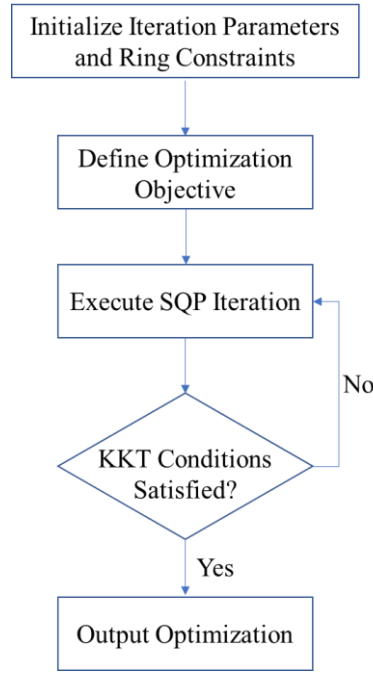


Fig. 4 Optimization Flowchart Based on the SQP Algorithm

The flowchart in **Fig.4** illustrates how the above equations are optimized via programming. Naturally, the precise optimization objectives and the choice of constraint strategies depend on the specific circumstances.

3. Case Studies And Discussions

Deploying multiple actuators at different positions can alleviate stress concentration. The parameters of the polypropylene ring are detailed in **Table 1**. To illustrate the effectiveness of multi-faceted actuation and control strategies, we conducted systematic tests and comparisons.

Table 1. Parameters and properties of the ring

Property	Values
Ring radius R , m	0.05
Ring width b , m	0.010
Ring thickness h , m	0.001
Young's modulus of ring Y , N/m ²	1.556×10^9
Ring mass density ρ , kg/m ³	907
Poisson's ratio μ	0.3
Flexoelectric patches thickness h^a , μm	50
Flexoelectric constant π_{12} , $\mu\text{C/m}$	100
AFM probe tip radius r , nm	50
Actuation voltage ϕ^a , V	10

3.1. Key factors that affect flexoelectric actuation effect

3.1.1 Parametric study on actuator characteristics and their impact on structural displacement

Figure 5 and **Figure 6** clearly show the relationship between the displacement at the ring's zero point, which means zero initial point in polar coordinates, and the placement of the actuators, with the polar

plots revealing a sinusoidal distribution. It can also be seen that as the mode number increases, the amplitude of the curve decreases.

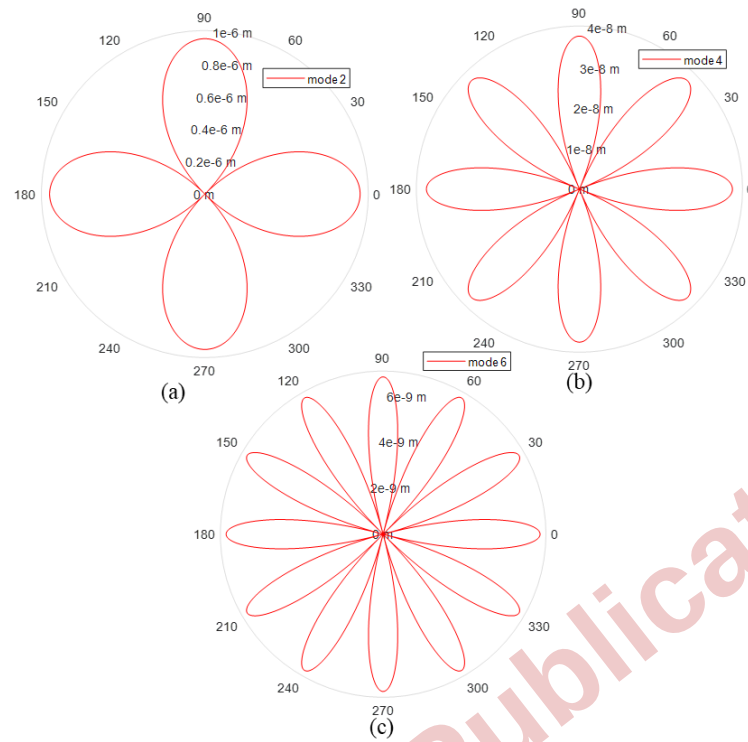


Fig.5 Analysis of even mode displacement of circular ring with respect to actuator positions: (a) mode 2, (b) mode 4, (c) mode 6.

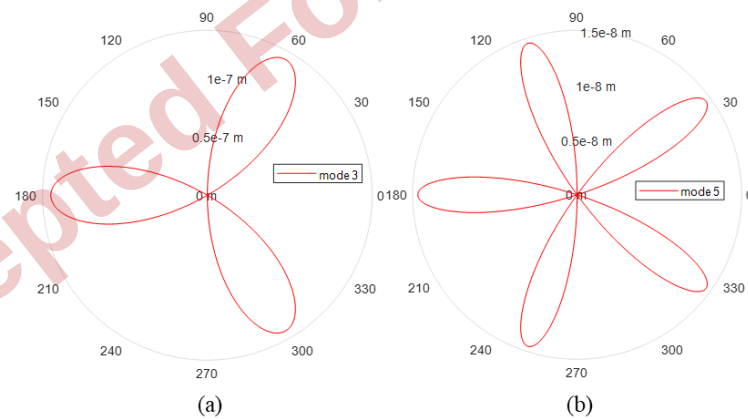


Fig.6 Analysis of odd mode displacement of circular ring with respect to actuator positions: (a) mode 3, (b) mode 5.

The images are divided into two groups: even-mode and odd-mode. Observations indicate that the distribution of actuators does not significantly alter the symmetry of the ring. For example, numerical calculations for the mode 2 distribution show an effect of approximately 10^{-9} in magnitude, which has a slight impact on subsequent control. Additionally, the distribution in odd-mode cases exhibits some peculiarities compared to even-mode ones. In odd-mode distributions, there are only corresponding numbers of branches. This is because the symmetry disruption is not significant, causing the displacement at relatively positioned points to be almost equal in absolute value, meaning branches at angles differing by 180° overlap.

Figure 7 clearly demonstrate a linear relationship between the displacement at the ring's zero point and the applied excitation voltage, more specifically a proportional relationship, where changes in

excitation voltage directly influence the degree of displacement. This observed linear relationship emphasizes the system's responsiveness to changes in applied voltage.

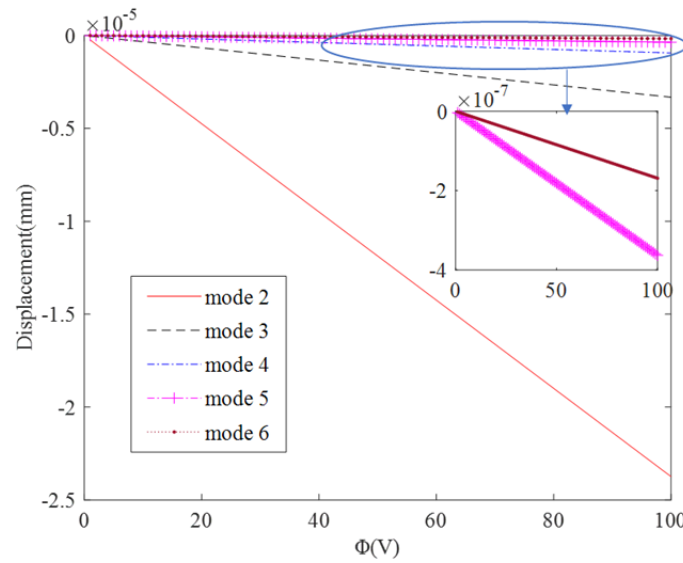


Fig.7 Analysis of displacement of zero points in a ring with respect to actuator driving voltage

Close examination of the change rates of these lines reveals that the mode 2, exhibits the steepest increase, indicating the highest sensitivity to changes in excitation voltage. Subsequently, the change rate decreases sequentially with each higher mode, with the mode 5, or mode 6, showing the slowest displacement response to changes in excitation voltage. This layered arrangement of change rates among the models provides quantitative insights into the relative strength of flexoelectric effects within each system.

The trend of displacement with changing flexoelectric patch thickness is almost consistent across different modes. Therefore, for the sake of research convenience, we chose mode 2 as the research target. **Fig.8** shows a consistent trend in the relationship graphs about mode 2 under different voltages. For clarity, we first focused on the relationship between the displacement at the ring's zero point and the flexoelectric patch in the mode 2. Close observation shows that when the flexoelectric patch thickness is less than 1 micron, the displacement at the ring's zero point increases rapidly with an increase in patch thickness. However, once the thickness exceeds 1 micron, further increases in patch thickness result in negligible changes in displacement. This phenomenon can be explained by the fact that the electric field gradient along the radius direction on the flexoelectric patch is maximal on the patch's surface but rapidly decreases towards the ring's center until it can be considered zero.

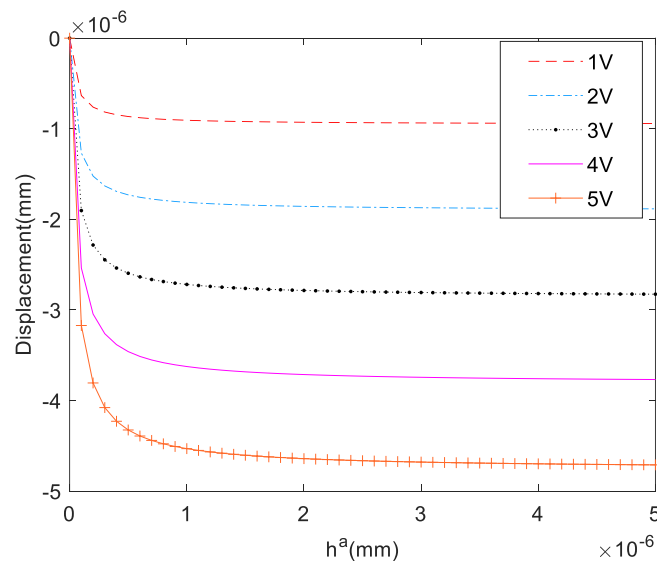


Fig.8 Analysis of even-mode displacement of zero points in a circular ring with respect to actuator thickness

Additionally, it is worth noting that in practical engineering applications, the thickness of flexoelectric patches is generally not less than 10 microns. Therefore, we believe that in the context of real-world engineering applications, changes in the thickness of the flexoelectric patches are considered insignificant to the overall displacement of the ring.

3.1.2 Analysis of effect of ring shell geometrical properties on vibration response

Figure 9 shows the trend of changes in ring radius and displacement magnitude in the mode 2 under different voltages. Similar to previous research cases, the trend of changes in ring radius and displacement magnitude is almost consistent across different modes. As seen from the trend in the figure, when the ring radius exceeds 0.1 meters, the displacement caused by flexoelectric actuation can be considered negligible. The same conclusion applies to other modes. This can be explained by the fact that as the radius continues to increase while the moment generated by the flexoelectric patches remains constant, the displacement will continuously decrease.

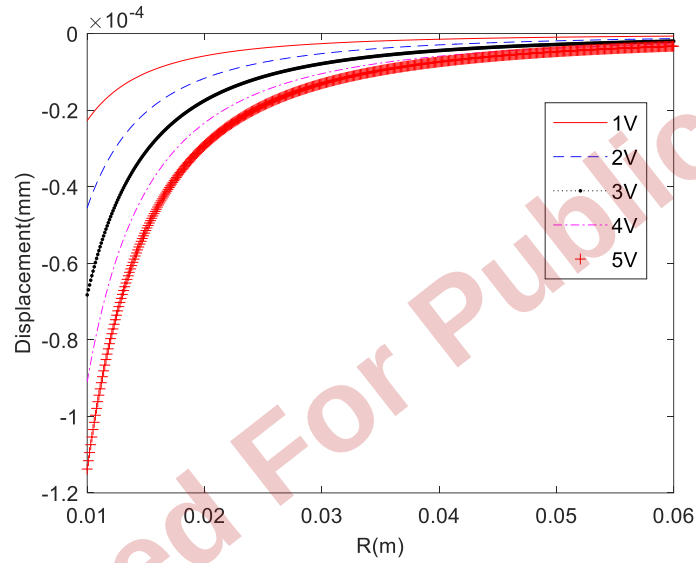


Fig.9 Analysis of even-mode displacement of zero points in a circular ring with respect to circular ring radius

Based on the above analysis results and modal comprehensive consideration, selecting a ring radius of 0.05 meters can achieve a relatively ideal effect in general situations.

Figure 10 shows the trend of changes in ring thickness and displacement in the mode 2 under different voltage amplitudes. Similar to previous studies, the trend of changes is almost consistent across different modes. Upon closer examination, it is observed that when the ring thickness exceeds 0.01 meters, the zero displacement of the ring can be considered negligible from an engineering perspective. This can be explained by the fact that as the ring thickness increases, the section modulus also increases, thereby reducing the displacement caused by the moment generated by the flexoelectric patches.

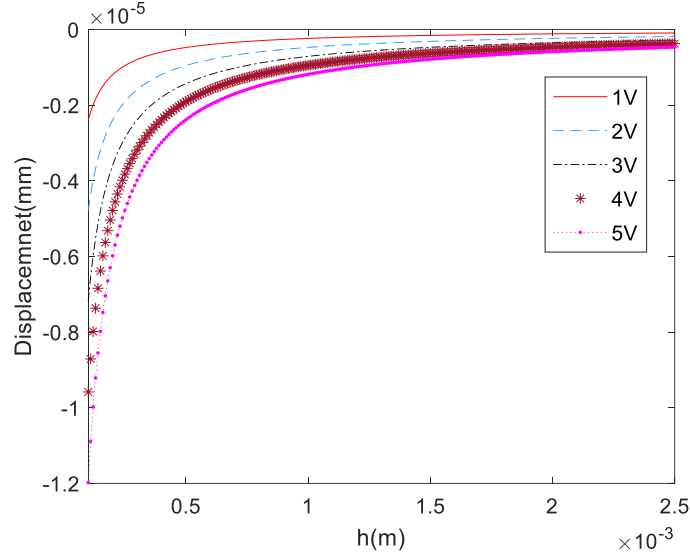


Fig.10 Analysis of even-mode displacement of zero points in a circular ring with respect to circular ring thickness.

In conclusion, for the final effect, selecting a ring thickness of 0.001 meters can achieve very significant actuation performance. Furthermore, considering maintaining a constant ring radius, if the ring thickness continues to increase, the structure will gradually evolve into a plate, so the ring thickness should not be excessively large.

3.1.3 Comprehensive evaluation of parameter effects

Based on the above analysis, it can be concluded that variations in ring radius, ring thickness, and flexoelectric patch thickness have limited influence on displacement within practical operating ranges. Therefore, for the sake of consistency and convenience in subsequent analysis, these parameters are fixed as follows: ring radius of 0.05 m, ring thickness of 0.001 m, and flexoelectric patch thickness of 50 μm . In addition, the effect of actuation voltage on displacement exhibits a linear relationship, making it straightforward to handle. Finally, the actuator position has a significant and more complex impact on structural response, which will be explored in detail in the following sections.

3.2. Optimal actuated position analysis

Considering the ring's actuation in a zero-external-force floating state, the roundness function $\Gamma(\mathbf{x}) = R_i / R_c$ is minimized via the SQP algorithm, which rapidly computes both $\min[\Gamma(\mathbf{x})]$ (i.e., the optimal actuation performance) and the corresponding value of its independent variable. The output thus directly identifies the optimal sensor location.

The reason we seek the minimum of the function $\Gamma(\mathbf{x}) = R_i / R_c$ is that, by its definition, a smaller $\Gamma(\mathbf{x}) = R_i / R_c$ value corresponds to a greater deformation, and thus a more pronounced actuation effect. After computing the optimal value, we randomly select several other actuator configurations for comparison. The results likewise confirm the validity of our optimization, as shown in **Fig.11** and **Fig.12**, where we illustrate the two-actuator case across different modes.

Below, specifically **Fig.13** and **Fig.14**, are the optimal results filtered through the optimization algorithm and the displacement distribution table produced by different numbers of actuators driving the ring structure. Detailed data can be found in **Tables 2–5** in Appendix A.

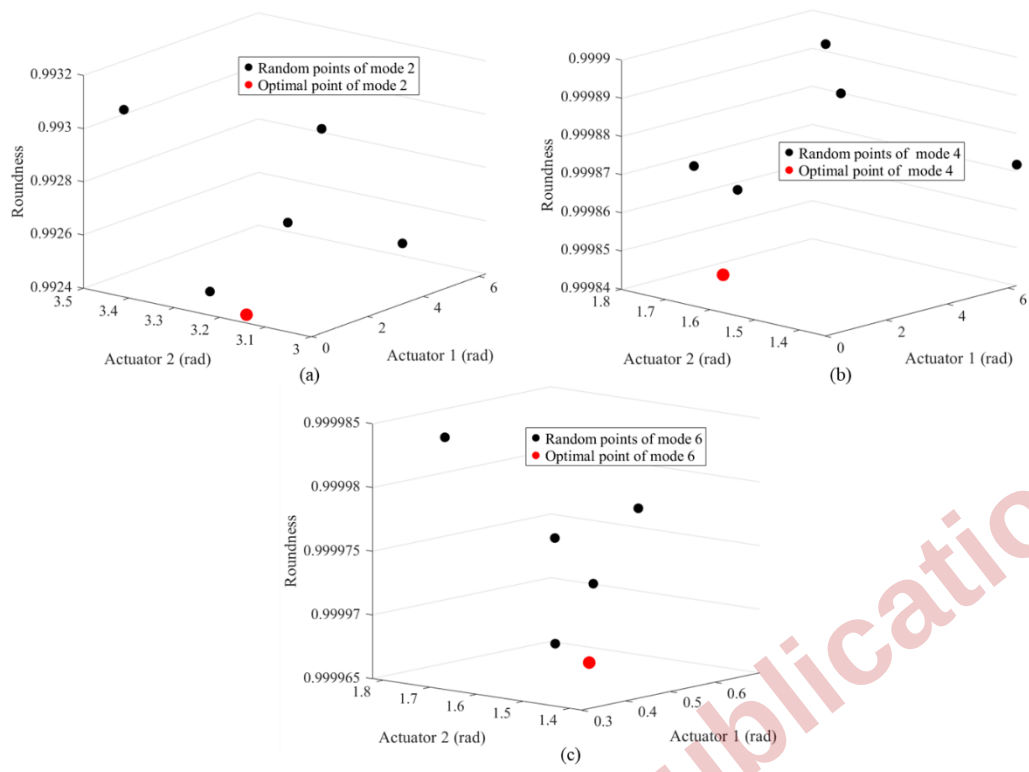


Fig. 11 Comparison of Roundness at the Optimal Actuation Positions versus Randomly Generated Points for even mode: (a) mode 2, (b) mode 4, (c) mode 6.

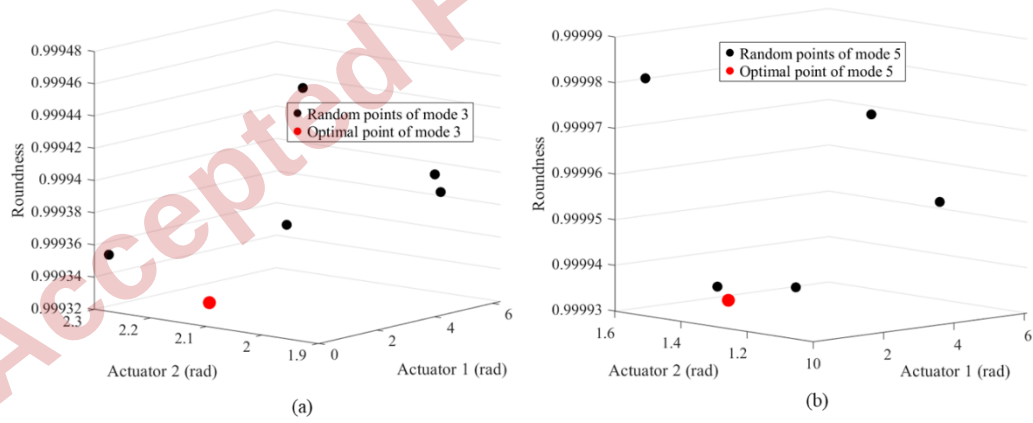


Fig. 12 Comparison of Roundness at the Optimal Actuation Positions versus Randomly Generated Points for odd mode: (a) mode 3, (b) mode 5.

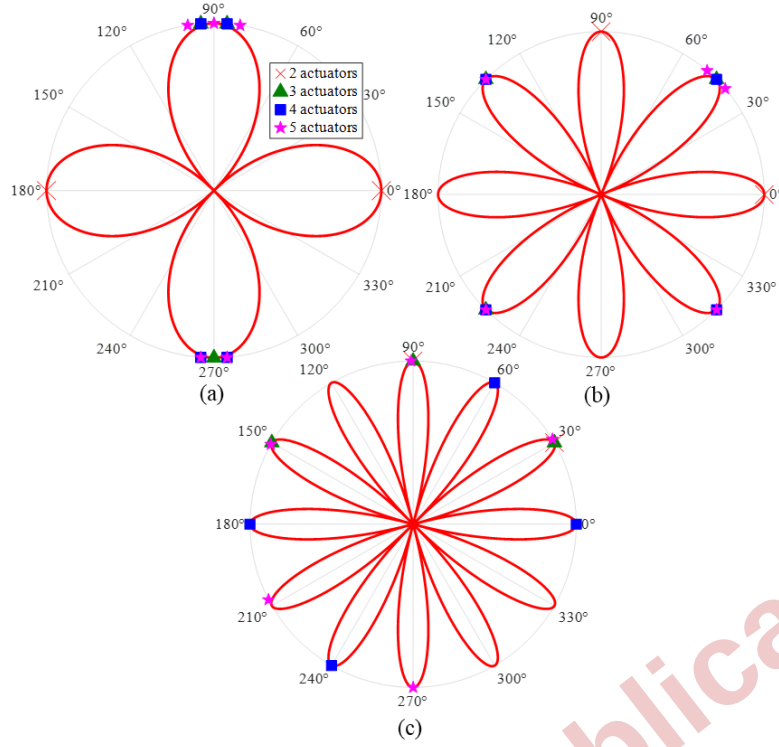


Fig. 13 Optimal actuation positions for even mode with varying numbers of actuators: (a) mode 2, (b) mode 4, (c) mode 6.

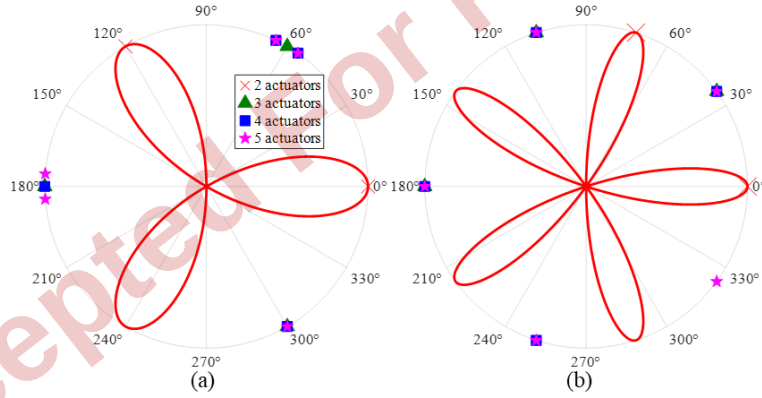


Fig. 14 Optimal actuation positions for odd mode with varying numbers of actuators: (a) mode 3, (b) mode 5.

3.3. Optimal controlled position analysis

Similarly, considering to apply a step force F of magnitude 10 N at the point where $\psi = 0$, directed toward the center of the circle, optimal control is achieved when the overall vibration of the ring is cancelled; in other words, by finding $\Gamma(\mathbf{x}) = R_i / R_c = 0$, which means find the set of $Z_f = \{\mathbf{x} \in \mathbb{R} \mid \Gamma(\mathbf{x}) = 0\}$. The root of the $\Gamma(\mathbf{x}) = R_i / R_c$ can likewise be determined rapidly via the SQP algorithm—corresponding to optimal control performance—as well as the associated values of its independent variables.

Below, specifically **Fig.15** and **Fig.16**, we continuously obtained the optimal control position distribution results from two to five actuators. Given the large number of different sensor position combinations, traditional exhaustive methods have low computational efficiency. In contrast, the SQP iterative method effectively solves this problem. The SQP algorithm, similar to an adaptive gradient descent algorithm, requires only the identification of potential optimal positions before calculations and

can quickly yield accurate results once substituted. By strategically placing the sensors, the sensor signals can be significantly enhanced, enabling precise monitoring of structure vibrations dominated by bending deformation, which can be proved from the below tables. Detailed data can be found in **Tables 6–9** in Appendix A.

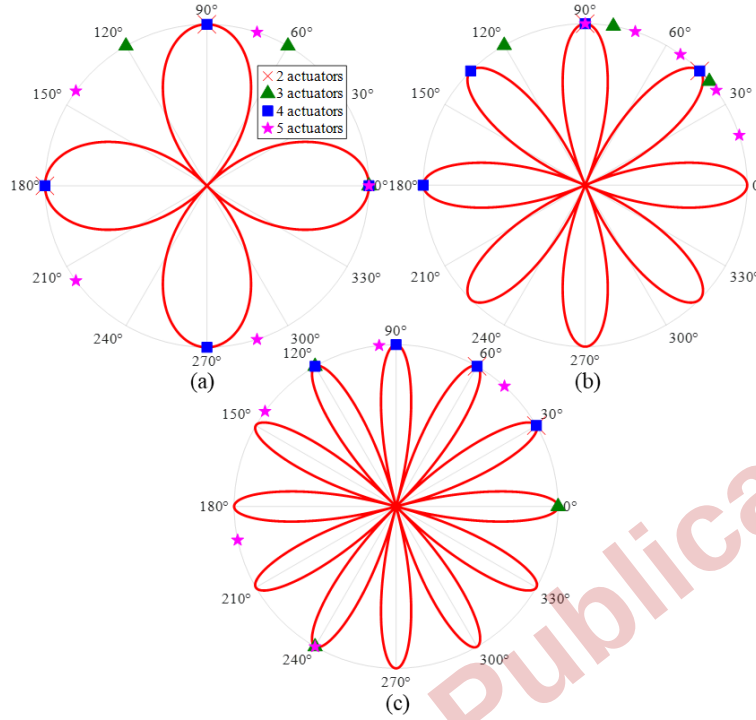


Fig.15 Optimal control positions for even mode with varying numbers of actuators: (a) mode 2, (b) mode 4, (c) mode 6.

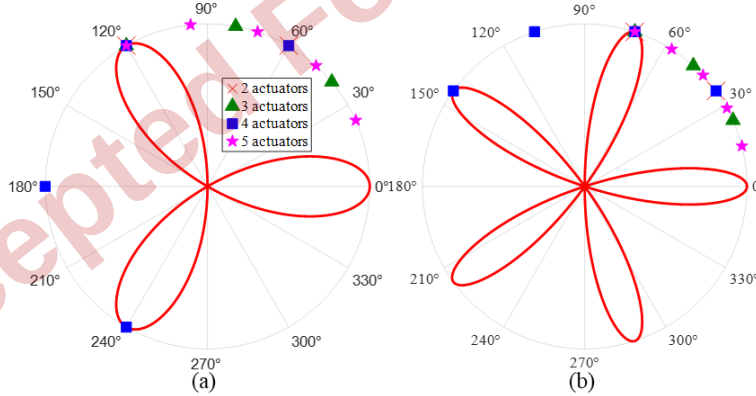


Fig.16 Optimal control positions for odd mode with varying numbers of actuators: (a) mode 3, (b) mode 5.

Summarizing and analyzing the data above, it can be found that for the actuator distribution in a particular mode, if studied in a polar coordinate system, the relationships between these points satisfy the cyclotomic polynomial. The distribution follows the rule: $\varepsilon_i = \cos[(2i\pi)/k] + j\sin[(2i\pi)/k]$, where ε are the roots of the polynomial $\phi_n(x) = \prod_{\substack{1 \leq i \leq k \\ \gcd(i,k)=1}} (x - e^{(j*2\pi i)/k})$. By the properties of cyclotomic polynomials, $\sum_{i=1}^k \cos[(2i\pi)/k] = 0$, $\sum_{i=1}^k \sin[(2i\pi)/k] = 0$, which means $\sum_{i=1}^k \varepsilon_i = 0$. This means the sum of displacements caused by all actuators is zero, consistent with the calculations.

From the above analysis, if the overall superimposed displacement of the ring structure under multiple actuators is to be zero, their distribution only needs to satisfy the cyclotomic polynomial condition. It is also worth noting that the properties of the cyclotomic polynomial apply to all distributions cases for $k \geq 2$.

3.4. Analysis of the appropriate number of actuators

As shown in Fig.17, when the number of actuators is less than or equal to 5, the relationship between the maximum displacement of the ring and the number of actuators in all modes appears to be almost entirely linear. This phenomenon can be explained from two perspectives. First, the width of the flexoelectric sheets chosen in this study is relatively small at $\pi/20$, making the displacement change insensitive to the movement of the flexoelectric sheets near the peak. Second, as the mode of the ring increases, the number of branches at the peak increases, as shown in Fig.5 and Fig.6, resulting in more positions to choose from. Even the second mode has two maximum peaks and two minimum peaks. However, as the number of actuators continues to increase, the nonlinear relationship becomes more apparent.

However, having more and more actuators is not necessarily better. According to Eq. (23), as the number of actuators increases, the iteration matrix becomes more complex, leading to longer computation times.

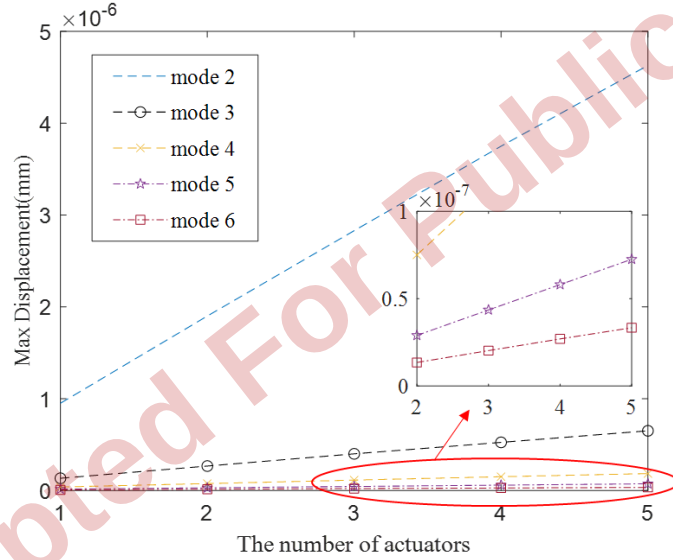


Fig.17 Distribution of the relationship between the maximum displacement of the ring and the number of actuators in different modes

4. Conclusion

This article delves into the driving and control of ring structures using multiple flexoelectric actuators. Considering the stress concentration induced by flexoelectric actuators, placing multiple actuators at optimal positions can enhance the driving performance and control reliability of the flexoelectric control system. An AFM probe was used to induce changes in the electric field gradient, generating concentrated stress in the flexoelectric actuators through the inverse flexoelectric effect. Subsequently, numerical simulations and analyses are conducted to study the impact of different parameters on the vibration characteristics of the ring, and corresponding optimization schemes were proposed.

In the analysis of actuator parameters and structural parameters, the influence of these parameters on the vibration characteristics of the ring was revealed by examining the displacement at the ring's zero point in relation to actuator position, driving voltage, thickness, as well as the ring's radius and thickness. Particularly, by analyzing the impact of actuator position on the symmetry of the ring, the differences between odd-order and even-order actuator distributions were discussed, providing a basis for subsequent

control scheme formulation.

The optimization section, based on the SQP method, proposed corresponding optimization method for external force driving and control. By defining a roundness function and utilizing the SQP algorithm, the optimal driving and control positions under different actuator conditions under external force were obtained. The results showed that the control effect was more significant under multiple actuators, and the distribution of control positions satisfied the cyclotomic polynomial, providing an effective strategy for vibration control of the ring structure.

Finally, the data also indicate that having more actuators is not necessarily better and should be controlled within a certain range. This study provides new ideas and methods for improving the performance and stability of micro-nano structure devices and for applying the flexoelectric effect in the micro-nano structure technology field.

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Data availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare no competing interests.

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Appendix A

Table 2. The observed optimal positions actuated by two actuators

Mode	Actuator 1, rad	Actuator 2, rad
2	0	3.1416
3	0	2.0944
4	0	1.5708
5	0	1.2566
6	0.5236	1.5708

Table 3. The observed optimal positions actuated by three actuators

Mode	Actuator 1, rad	Actuator 2, rad	Actuator 3, rad
2	1.4923	1.6493	4.7124
3	1.0472	3.1416	5.2360
4	0.7854	2.3562	3.9270
5	0.6283	1.8850	3.1416
6	0.5236	1.5708	2.6180

Table 4. The observed optimal positions actuated by four actuators

Mode	Actuator 1, rad	Actuator 2, rad	Actuator 3, rad	Actuator 4, rad
2	1.4923	1.6493	4.6338	4.7909
3	0.9687	1.1257	3.1416	5.2360
4	0.7854	2.3562	3.9270	5.4978
5	0.6283	1.8850	3.1416	4.3982
6	0	1.0472	3.1416	4.1888

Table 5. The observed optimal positions actuated by five actuators

Mode	Actuator 1, rad	Actuator 2, rad	Actuator 3, rad	Actuator 4, rad	Actuator 5, rad
2	1.4137	1.5708	1.7279	4.6338	4.7909
3	0.9687	1.1257	3.0630	3.2201	5.2360
4	0.7069	0.8639	2.3562	3.9270	5.4978
5	0.6283	1.8850	3.1416	4.3982	5.6549
6	0.5482	1.5831	2.6303	3.6229	4.7124

Table 6. The observed optimal positions controlled by two actuators

Mode	Actuator 1, rad	Actuator 2, rad
2	1.5708	3.1416
3	1.0471	2.0944
4	0.7854	1.5708
5	0.6283	1.2566
6	0.5236	1.0472

Table 7. The observed optimal positions controlled by three actuators

Mode	Actuator 1, rad	Actuator 2, rad	Actuator 3, rad
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2	0	1.0471	2.0944
3	0.6981	1.3963	2.0944
4	0.5236	1.0471	1.5708
5	0.4189	0.8376	1.2566
6	0	2.0944	4.1888

Table 8. The observed optimal positions controlled by four actuators

Mode	Actuator 1, rad	Actuator 2, rad	Actuator 3, rad	Actuator 4, rad
2	0	1.5708	3.1416	4.7124
3	1.0472	2.0944	3.1416	4.1888
4	0.7854	1.5708	2.3562	3.1416
5	0.6283	1.2566	1.8850	2.5133
6	0.5236	1.0472	1.5708	2.0944

Table 9. The observed optimal positions controlled by five actuators

Mode	Actuator 1, rad	Actuator 2, rad	Actuator 3, rad	Actuator 4, rad	Actuator 5, rad
2	0	1.2566	2.5133	3.7699	5.0265
3	0.4189	0.8378	1.2566	1.6755	2.0944
4	0.3142	0.6283	0.9425	1.2566	1.5708
5	0.2513	0.5027	0.7540	1.0053	1.2566
6	0.8378	1.6755	2.5133	3.3510	4.1888